CLINICAL SAFETY MANUAL & BIOMECHANICAL BOUNDARIES

Section 1 Constraints, Section 5 Progression Rules & Rapid Safety Check

MANDATORY CLINICAL RESTRICTIONS:

• **Cervical Spine (C4–C7 Fusion)**: Strict neutral alignment. Zero forced end-range extension or neck rotation. No classical Pilates head curls (head grounded).
• **Right Shoulder (Subacromial/C7–T1)**: Active elevation strictly limited below 80°–90°. Elbows tucked below shoulder line during all rows and presses.
• **Lower Chain (Left Knee Osteotomy & Right Ankle)**: Braced on land. Max 45° shallow knee bend. Zero crouching or kneeling directly on unpadded surfaces.

1. Multi-Disciplinary Clinical Constraints

• **Cervical Spine (C4–C7 Fusion & C7–T1 Junction)**: Strictly preserve neutral cervical alignment. No forced end-range extension or high-velocity rotation. Foam rolling the neck or lower back is prohibited. Traditional Pilates head/shoulder curls (e.g. classical Hundred) are contraindicated; use supported preparation with head grounded.

• **Right Shoulder (Frozen Shoulder & Nerve Root Irritation)**: Limit active elevation below 80°–90°. All banded rowing and aquatic presses must keep elbows tucked below shoulder height to prevent subacromial impingement and C7–T1 neuropathic burning.

• **Lower Kinetic Chain (Left Knee Post-Osteotomy & Right Ankle Instability)**: Braced during loaded land exercises. No deep squatting, crouching, or kneeling on hard unpadded surfaces. Aquatic immersion (chest-depth) offloads 60–75% of compressive joint loading, allowing high concentric intensity without eccentric cartilage breakdown.

• **Hydrotherapy Thermal Environment**: Hydrotherapy combines warm-water buoyancy (30°C–34°C) with hydrostatic pressure and fluid resistance. Using heavy Therabands and pool rails allows intense muscular loading with near-zero spinal axial compression.

2. Clinical Progression & Regression Rules (Section 5)

• **Progression Criteria**: Advance band resistance only when a routine is completed with post-exercise pain delta >= 0 (no pain increase) for 3 consecutive sessions and baseline pain is < 5/10.

• **Regression Criteria**: If sharp C7–T1 burning nerve pain or acute knee/ankle instability occurs, immediately regress to passive decompression (Hydro H-MED1 or Yoga Y-BEG1) and apply cryotherapy.

• **Equipment Anchor Safety**: Always verify anchor points for Therabands in the pool. Never secure bands to loose fittings.

RAPID CLINICAL SAFETY REFERENCE: DO / DON'T GUIDE

Always cross-check daily exercise plans against these mandatory limits.

CLINICAL DOMAIN	MANDATORY 'DO' (APPROVED)	STRICT 'DON'T' (CONTRAINDICATED)
Cervical Spine (C4–C7)	• Maintain neutral cervical alignment • Head grounded on mat with thin pad • Micro-chin tucks within pain-free arc	• NO neck extension or rapid rotation • NO foam rolling neck or lower back • NO classical Pilates head curls/Hundred
Right Shoulder & C7–T1	• Keep elbows tucked below shoulder line • Active elevation strictly below 80°–90° • Gentle non-tension nerve flossing	• NO overhead reach >90° • NO high-row elbow flaring • Stop instantly if neuropathic burning occurs
Lower Chain (Knee/Ankle)	• Wear knee/ankle brace on loaded land work • Max 45° knee flexion on mini-squats • Water immersion offloads 60–75% load	• NO deep squats or crouching • NO kneeling on hard unpadded surfaces • NO rotational twisting at knee or ankle
Core & Spinal Mechanics	• Isometric abdominal wall bracing • Supported bridge holds with sacral block • Slow controlled heel slides	• NO lumbar hyperextension during kicks • NO lumbar flexion during knee drives • NO unsupported heavy spinal loading
Equipment & Progression	• Anchor Therabands to solid pool rails only • Advance bands only after 3 zero-delta sessions • Use cushions under forearms and knees	• NEVER anchor bands to loose pool fittings • NEVER push through baseline pain >=5/10 • NO sudden jumps in band resistance

CLINICAL GOVERNANCE & SAFETY DISCLAIMERS

1. This is appointment preparation and second-opinion context, not a clinical recommendation to act on independently. Discuss with your physiotherapist and neurosurgeon before making changes to your treatment.

2. Any changes to medication or pharmacological therapies must be reviewed, prescribed, and managed by your General Practitioner or managing specialist in consultation with a qualified pharmacist.

3. If you experience sudden severe motor weakness, progressive numbness, fever with joint swelling, or acute changes in bowel or bladder control, seek immediate emergency medical evaluation.